MASTER REFERENCE · READ THIS BEFORE THE GUIDES

Diagnostic reference package

Print-ready shop set for the stated vehicle: 2004 Jeep Grand Cherokee Laredo, 4.0L inline-six, described as the California emissions package with four oxygen sensors. This binder is the master reference. Guides 01 through 10 are the same pages, split so one job can be printed by itself.

CONFIRMED Code title, engine design, or a quoted bulletin. **CATALOG LISTED** Maker lists it. Still verify by VIN. **LIKELY** Fits this engine. Confirm on the truck. **POSSIBLE** Credible. Not proven here. **TEST REQUIRED** Measure before you buy. **NOT CONFIRMED** Checked and not supported. Do not buy from it.

Three problems. Keep them apart.

Idle chain POSSIBLE

Pintle moved by hand → IAC may be damaged or held open → extra air → high or unstable idle → P0505 may set → mixture may go lean → manifold may glow. A vacuum leak is a likely idle cause on this MAP engine and only a possible cause of a lean glow. A misfire can glow the iron without this chain.

Oil and fire SEPARATE

Oil leak → oil on the hot manifold, downpipe, or an oxygen sensor → fire hazard. Independent of the IAC. Fix the leak before you replace a sensor that was sitting in the oil. Trace fresh oil. Do not assume a rear main.

P0161 SEPARATE

Heater circuit, Bank 2 Sensor 2. Rear downstream sensor after the rear mini-cat. Test connector, fuse, cold heater resistance, 12 V feed, and PCM control. The code does not prove a lean engine or a bad sensing element.

What this package will not do

- It will not turn the working theory into a diagnosis. The test record in Guide 01 is where readings go.
- It will not give a fuse number or a wire color. Those vary, and a wrong color sends the test to the wrong cavity. Use the under-hood lid and the resistance test to identify the heater pair.
- It will not approve carb-cleaner or propane leak spray while oil is on the exhaust or the manifold has been glowing. Smoke and a visual inspection are the tests.
- It is not the factory service manual. Torque values and part numbers are labeled with how they were sourced. Verify by VIN.

Safety overrides the order of the other tests. Glowing manifold or oil reaching hot exhaust: stop, cool completely, Guides 05 and 06. If the glow returns after a repair, stop immediately and tow. Do not clear codes and drive it.

MASTER REFERENCE · CONTENTS

What to print

Guide	Use it for	Pages
00 Master reference	This cover, the evidence rules, and the whole binder in one PDF.	Binder
01 Diagnostic guide	Vehicle identification, four-sensor layout, both codes, the idle screen, the symptom table, the short decision path, and a blank test record.	6
02 P0161	Bank 2 Sensor 2 heater circuit. Connector, fuse, resistance, power, PCM control, and when a sensor is actually the part.	4
03 P0505 / IAC	Stepper operation, the do-not-touch-pintle rule, leak versus IAC, ohms, throttle-body cleaning, installation, PCM reset, and relearn.	4
04 Vacuum / EVAP	Why a leak raises idle on a MAP engine, where the brittle pieces are, safe tests, PCV/CCV and purge.	4
05 Glowing manifold	Safety stop. Possible causes. The lean-heat path written as possible. What continued running damages.	2
06 Oil / fire	Safety stop. Degrease and trace fresh oil. The five common 4.0L leak sources. Sensor replacement rule.	2
07 Repair priority	The thirteen-step order, from the safety stop through the rescan, including the stop-if-it-glows-again line.	2
08 Flowcharts	Full-page P0505 tree and full-page P0161 tree.	2
09 Parts	What checked out, what did not, and the test screens with their confidence labels.	2
10 One-page sheet	Color shop card: both codes, both warnings, the IAC rule, the test screens, and the thirteen steps.	1

Suggested shop use

Guide 10 on the vehicle. Guide 07 on the clipboard. Guide 05 or 06 in the bay if that hazard is present. The matching test guide open while the meter is in your hand. Guide 09 only when a test has already named a part. Guide 01 page 6 filled in before codes are cleared.

Files

Each guide is a letter-size PDF. The same pages are exported as PNG and JPEG for a phone or a tablet in the bay. Print the PDFs in color. The red, orange, blue, and green boxes are the priority code, not decoration.

Parts that must not be bought from memory. NTK 23142 and Mopar 56029001AB did not confirm as the 2004 Grand Cherokee 4.0L Bank 2 Sensor 2. Part 53030006 did not confirm as a vacuum elbow for this Jeep. The numbers that did catalog-list are in Guide 09, and they still get a VIN check. Dorman 47128 / 4854265 is a short CCV hose set, not every vacuum line. The IAC candidate is a new 4874373AB.

Package date 12 October 2026. Sources used for the labeled facts include the OBD-II definitions of P0161 and P0505, Chrysler TSB 18-008-04, the 2004 WJ HVAC vacuum rapid-response 04-008, the JeepSpecs WJ four-sensor layout and 2004 part list, Dorman's listing for 47128, and a Mopar catalog listing for 4874373AB. Earlier 4.0L heater-ohm and heater-volt procedures are cited as related screens, not as this VIN's specification page. Where those sources disagree with a sticker or a VIN lookup, the vehicle wins.

Master vehicle diagnostic guide

Shop reference for this truck. It separates three problems that can be present at the same time. A working theory is not a completed diagnosis. Every claim below carries a label.

CONFIRMED Design fact, standard code title, or a quoted factory bulletin. **CATALOG LISTED** Listed by the maker for this model. Still verify by VIN. **LIKELY** Fits the engine and the published layout. Confirm on the truck. **POSSIBLE** A credible cause. Not proven on this vehicle. **TEST REQUIRED** Do not replace a part until this measurement is made. **NOT CONFIRMED** Number or location that did not check out. Do not buy from it.

Vehicle identification

Item	What to use	Label
Vehicle	2004 Jeep Grand Cherokee, WJ, Laredo, as stated for this job. Read the VIN on the dash and the door sticker before ordering parts.	STATED
Engine	4.0L inline-six, 242 cu in. Chrysler sales code ERH. The 8th VIN character is usually S. Read it. Do not assume from the badge.	LIKELY
Air metering	Speed-density. The PCM uses the MAP sensor, RPM, and intake-air temperature. There is no mass-airflow sensor. A vacuum leak is not automatically "unmetered air" the way it is on a MAF engine.	CONFIRMED
Throttle and idle	Cable throttle. A 4-wire stepper IAC on the throttle body bypasses the closed plate. This is not drive-by-wire. Do not use an electronic-throttle relearn procedure.	CONFIRMED
Ignition	Coil rail, PCM-controlled. No distributor. Timing is not adjusted with a wrench.	CONFIRMED
Emissions	Described as the California package: four oxygen sensors, two mini-catalysts, and a main catalyst. Federal 4.0L trucks of this year use two sensors. Count the sensors and the mini-cats before you buy anything. Where the truck was purchased does not decide the package.	TEST REQUIRED
Transmission	Not assumed. Idle in Drive is a different load than idle in Park. Write down the range for every RPM reading.	TEST REQUIRED

Stop driving if either is true

The exhaust manifold is glowing orange or red, or engine oil is reaching the hot manifold, a downpipe, or an oxygen sensor. Cool the engine completely. Guides 05 and 06. Do not idle it "to see."

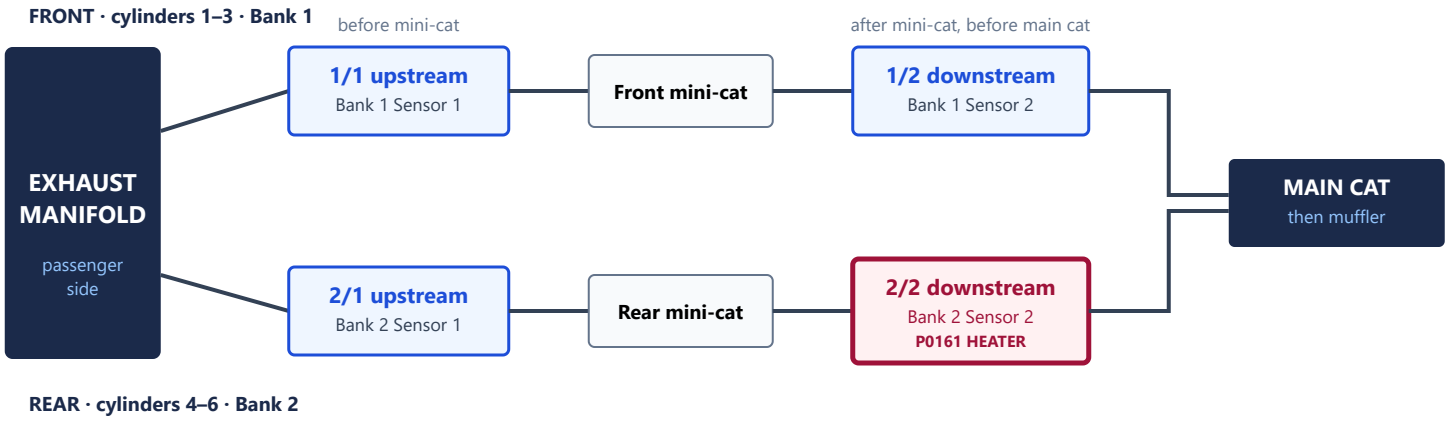
Print and use in this order

One-page sheet on the vehicle. Priority sheet next. Then the guide for the fault you are testing. Parts sheet last, and only after a test points at a part.

MASTER VEHICLE DIAGNOSTIC GUIDE · SENSOR LAYOUT

Four-sensor California layout

Published WJ layout for the 4.0L California package. **LIKELY** Confirm every bung on this truck before you unplug a sensor. Front of the engine is the radiator end. Cylinder 1 is at the front.



How the banks are named

Sensor 1 is upstream of that branch's mini-cat. Sensor 2 is downstream of the mini-cat and still ahead of the main cat. Bank 1 follows the front three cylinders. Bank 2 follows the rear three. **LIKELY** The harness length and the connector are different at each bung. Do not swap sensors as a test.

P0161 is not "the rear manifold sensor"

It is the heater circuit of Bank 2 Sensor 2: the rear downstream sensor, after the rear mini-cat. The published guide puts these sensors in the downpipes, not in the cast manifold. Trace the rear pipe.

TEST REQUIRED

Source of the four-position description: compiled WJ oxygen-sensor guide (JeepSpecs), consistent with repeated service descriptions of two mini-cats ahead of a main cat. It is not a page from the factory manual in hand. If this truck has only two sensors, stop and treat it as the federal package.

P0161 and P0505

P0161

CONFIRMED OBD-II title: O2 sensor heater circuit, Bank 2 Sensor 2.

The monitor is about the heater circuit. It is not a measurement of mixture, catalyst efficiency, or manifold temperature.

The circuit can fail because of: an open or shorted heater element, a corroded or oil-filled connector, a broken or melted wire, a fuse or ASD-relay feed, or the PCM control that grounds the heater. **POSSIBLE** each one. **TEST REQUIRED** before a sensor is bought.

This code does not prove the sensing element is dead, the engine is lean, or this sensor made the manifold glow. Oil on a connector can cause it. Oil on the sensor body more often ruins the signal than the heater. Test, then decide.

P0505

CONFIRMED Generic OBD-II title: idle control system malfunction.

On this Chrysler family the same code is often described as an **IAC motor circuit** fault: a short to ground or to battery on one of the four driver circuits while the motor is being commanded. That description comes from Chrysler-oriented service explanations, not from a 2004 FSM page photographed for this package. **LIKELY** as the electrical meaning. Read the wording on a Chrysler-capable scan tool.

A high idle is a separate observation. It may be stored beside this code. A related description, P1294, is “target idle not reached” on many Chrysler calibrations. If the tool offers it, record it. **TEST REQUIRED**

P0505 does not prove a vacuum leak. A leak can still be the reason the idle is high. Check both the circuit and the air leaks. Guide 03 and Guide 04.

Normal warm idle

CONFIRMED Chrysler TSB 18-008-04, domestic 2004 Grand Cherokee, 4.0L sales code ERH, built before 6 March 2004: the revised PCM software sets warm idle at **about 744 rpm**. That bulletin does not prove this VIN already has that software.

Screening band for this job: about 700–800 rpm, engine fully warm, Park, parking brake set, A/C and other loads off, after the idle relearn. Use a scan-tool RPM if you have one. In Drive, with the brake held, expect the PCM to hold a similar controlled idle, not a flare. Above about 1,000 rpm with no load, after a real relearn, is not a normal result. **TEST REQUIRED** Write down coolant temperature, gear, and whether the A/C is on. A cold engine idles higher on purpose. Do not judge the IAC until the engine is warm.

TSB source text: the new software “sets the engine idle speed at approximately 744 rpm (engine fully warm).” Subject of that bulletin was a low-idle drone and, on some Selec-Trac trucks, a transfer-case lamp. It is an idle-speed reference, not a diagnosis of this truck.

Symptoms and what they do not prove

What you see	Credible causes	Not proven by that symptom alone
Warm idle high or hunting	IAC pintle too far open, IAC O-ring leak, vacuum leak, throttle plate not closed, cable, carbon in the bore. POSSIBLE	A lean mixture. A glowing manifold. A failed O2 sensor.
P0505	IAC winding, connector, driver wire, or PCM driver. A pintle that was pushed can damage the motor electrically or mechanically. POSSIBLE	A vacuum leak. The leak can still be there. Ohms that look normal do not clear a pushed pintle.
P0161	Heater element, fuse, ASD feed, wiring, connector, PCM control. POSSIBLE	That the sensor signal is bad, or that this code caused a glow or a high idle.
Manifold glowing	Lean burn, misfire lighting fuel in the exhaust, retarded timing, restricted catalyst. POSSIBLE	That the IAC is the cause. Stop driving first. Guide 05.
Oil on the exhaust	Valve cover, oil-filter adapter, pressure sender, pan, rear main. POSSIBLE Trace fresh oil.	That the O2 heater failed. Fix the leak before a new sensor goes in.
Brown or purple manifold, seen cold	The iron has been very hot at some time. POSSIBLE past overheat.	That it is glowing now. An active glow is bright orange or red while hot. Stain alone is a clue, not a stop-driving event by itself.

Three problems. Do not collapse them into one.

Working theory POSSIBLE

Pintle was moved by hand → IAC may be damaged or held open → extra air → high or unstable idle → P0505 may set → mixture may go lean → exhaust may get hot enough to glow the manifold. Each arrow is unproven until the test beside it is done.

Separate hazard

Oil leak → oil on hot iron or on an O2 sensor → fire hazard. This chain does not depend on the IAC. It can destroy a new sensor and it can ignite if the manifold is hot. Guide 06.

Separate circuit

P0161 → Bank 2 Sensor 2 heater circuit → test connector, fuse, heater ohms, 12 V feed, and PCM control. Do not "fix" this code by replacing the IAC, and do not clear it and hope.

Why the lean step is only possible. This PCM is speed-density. Extra air from a manifold leak raises the MAP reading, and the PCM adds fuel for that reading. Idle still climbs, because the air is real. The mixture does not have to go lean. Prove a lean condition with upstream O2 voltage and fuel trims before you call the glow a lean-burn event. A misfire that dumps raw fuel into the pipe is an equally credible way to make iron glow.

MASTER VEHICLE DIAGNOSTIC GUIDE · DECISION PATH

Diagnostic decision tree

Short form. The full-page trees are Guide 08. Follow Guide 07 for the repair order. Do not start a road test while the manifold glows or oil is on the exhaust.

Start here — safety

Glowing manifold or oil on hot exhaust?
YES → stop, cool, Guides 05 and 06. NO → continue.

P0505, high idle, or a moved IAC pintle?

Measure warm idle. Screen 700–800 rpm in Park, loads off.
TSB reference about 744 rpm. TEST REQUIRED

High or unstable → smoke-test and inspect vacuum, CCV, EVAP, IAC O-ring, throttle plate.
Guide 04. Fix leaks before the relearn.

Pintle was pushed, or coils are open or shorted → new IAC. Do not touch the new pintle.
Guide 03. Ohms in range do not save a pushed motor.

Clean the throttle body. Plate must close. Cable must return.

PCM adaptive reset and idle relearn. Verify warm idle. Rescan.

P0161 — separate path

Identify Bank 2 Sensor 2 on the rear pipe, after the rear mini-cat.
TEST REQUIRED Count sensors first.

Connector, harness routing, oil in the plug, melted insulation.

Fuse and ASD feed. Use the under-hood lid diagram. Do not guess the cavity.

Heater resistance, sensor cold.
Screen about 4–10 Ω. Open = failed heater. TEST REQUIRED

About 10–12 V on the feed while the ASD is actually on, and PCM control on the other heater wire.

Repair the wire, fuse, or connector — or replace the sensor if the heater itself is open or shorted.

If oil is washing this sensor, fix the leak first or the new part dies the same way.

MASTER VEHICLE DIAGNOSTIC GUIDE · TEST RECORD

Test record — fill this in

Write the reading, not a conclusion, in the middle column. Judge it only after the guide for that test is followed. VIN last 8: _____ Mileage: _____ Coolant temp at idle check: _____

Test	Reading	Judgment / note
Sensor count (2 or 4) and mini-cat count		
Active manifold glow, or stain only, cold		
Oil reaching exhaust or a sensor? Fresh path from where?		
Warm RPM, Park, loads off		Screen 700–800. TSB ref. ~744
Warm RPM, Drive, brake held, loads off		Only if the vehicle is safe to put in gear
IAC coil A ohms / coil B ohms		Screen 40–80 Ω each. Isolated from each other
Was the pintle pushed or pulled by hand?		If yes, replace. Do not reuse.
Vacuum at warm idle, inHg, and does it flicker?		Typical screen about 17–21 inHg near this elevation
Smoke test: where smoke escaped		
Upstream fuel trims at idle and at 2,500 rpm		Record STFT and LTFT. Do not guess lean.
B2S2 heater ohms, cold, which cavities		Screen 4–10 Ω. OL = open heater
B2S2 heater volts, ASD on or engine running		Screen about 10–12 V
B2S2 control wire: does the PCM pull it?		Compare with another sensor if needed
Codes present before clearing, with freeze frame		
Codes after repair, idle verified		

Do not clear codes first. Freeze-frame RPM, coolant temperature, and fuel trims are the evidence. Photograph the scan tool. Clear only after the sheets above are filled and the repair that the tests support is done.

Elevation note: Tulsa is roughly 700 ft. A sea-level vacuum reading is a fair screen here. A low reading has more than one cause (leak, late timing, low compression, restricted exhaust). It is not a verdict.

P0161 OXYGEN-SENSOR HEATER GUIDE

P0161 — Bank 2 Sensor 2 heater circuit

CONFIRMED The OBD-II definition is a heater-circuit fault on Bank 2, Sensor 2. On the California 4.0L layout that position is the **rear downstream sensor**: after the rear mini-catalyst and before the main catalyst. **LIKELY** location. Count the sensors on this truck before you order a part. A two-sensor federal truck does not use this layout.

What the heater does

Each sensor has four circuits: heater power, heater control, signal return, and the sensor signal. The heater brings the zirconia element up to operating temperature. Until it is hot, the PCM cannot trust the signal. P0161 means the heater circuit failed the monitor. It does not grade the air-fuel signal.

What not to do first

Do not buy a sensor because the code is stored. Do not swap it with another sensor. Do not clear the code to "see if it comes back" before the tests below are written down. Do not unplug a sensor on a hot pipe. Exhaust iron stays burn-hot long after shutdown.

Work in this order

1. **Cold visual.** Engine cool. Find the rear downstream sensor by tracing the rear downpipe, not by guessing at the connector colors. Look at the connector shell, the latch, the terminals, and the first 12 inches of wire.
2. **Fuse and ASD feed.** Identify the ASD relay and its fuse from the under-hood power-distribution-center lid. Do not use a fuse number from memory.
3. **Heater resistance** on the sensor side, sensor at room temperature. Screening band about 4–10 Ω.
4. **Heater power,** about 10–12 V, measured while the feed is actually energized.
5. **PCM control** on the other heater wire. Repair the failed side. Replace the sensor only when the element itself is open or shorted, or the connector on the sensor pigtail is destroyed.

Oil changes the order

If oil is inside this connector or coating this sensor, the leak is a fire issue and a parts issue. Guide 06. Clean and repair the leak before a new sensor is installed. An oil-soaked connector can cause P0161 even when the heater element is intact. **POSSIBLE** **TEST REQUIRED**

Wire colors are not printed in this guide. Published colors for earlier 4.0L years do not reliably match every 2004 connector. Identify the heater pair by the resistance test, then mark the cavities. A factory diagram for this VIN is the authority if the connector is damaged or the pins are not obvious.

P0161 · CONNECTOR, FUSE, HEATER RESISTANCE

Connector, fuse, and resistance

Connector and wiring inspection

- Latch broken, or the plug seated but not locked.
- Green corrosion, oil, or water in the shell.
- Terminals backed out or spread open. A gentle tug on each wire.
- Insulation melted or brittle where the harness passed the pipe.
- Harness clips missing, so the wire rests on the manifold or the downpipe.

Pull on the connector body, never on the wire entering the sensor. Do not grease the plug. Do not solder the sensor leads. A twisted, taped splice next to the exhaust will fail again. If a wire is broken, repair it with a sealed splice and reroute it in the original clips, with slack for engine rock.

Fuse and ASD feed TEST REQUIRED

Heater power on this Chrysler family is fed from the automatic-shutdown (ASD) relay circuit, not from a constant key-on wire. Open the under-hood center and read the lid. Find the ASD relay and the fuse that feeds it.

1. Remove that fuse and look at the element. Also measure continuity across the fuse. A fuse can look intact and be open.
2. With the fuse installed, key on, check for voltage on both fuse legs during the moment the circuit is powered. One live leg and one dead leg means the fuse is open.
3. If every oxygen-sensor heater feed is dead, suspect the fuse, the ASD relay, or the ASD output. If the other sensors have power and this one does not, suspect the branch wire to Bank 2 Sensor 2.

Heater resistance — sensor cold TEST REQUIRED

Screening band for this job: about 4–10 Ω at room temperature, measured on the sensor connector, not through the engine harness. Earlier 4.0L factory tests call for roughly 4–7 Ω, and a 1997 Grand Cherokee test calls for about 4.5 Ω to 7 Ω. Those limits are LIKELY cousins of this sensor, not a 2004 specification sheet in hand. Resistance rises as the sensor gets hot. A hot reading outside the band does not condemn the part. Let it cool.

Reading, sensor side, cold	Meaning	Next move
About 4–10 Ω on one pair only	Heater element is electrically plausible. IN RANGE	Do not replace yet. Test power and control.
OL, or many thousands of ohms, on every pair	Open heater, or you are not on the sensor connector.	Recheck that you are on the sensor pigtail. If still open, the sensor heater has failed.
Near 0 Ω	Shorted heater element.	Sensor is bad. Check the fuse too, because a short can open it.
Two different pairs both read a few ohms	You may be bridging signal and heater, or the element is shorted internally.	Stop and sort the cavities before you condemn the PCM.

How to find the heater pair without a diagram: the heater is the only pair that reads a handful of ohms when cold. The signal terminals of a cold narrowband sensor do not. Write down which two cavities those are. The engine-harness side is where voltage is tested, and it must mate to those same two cavities.

P0161 · POWER AND PCM CONTROL

Heater power and PCM control

Key-on is not always "circuit powered"
A steady key-on, engine-off reading of 0 V can be a false failure. On this Chrysler family the ASD relay often pulls in for about a second at key-on, then drops out if the engine is not turning. Earlier 4.0L procedures do specify about **10–12 V, key on, engine off**, on the heater feed. Use that number. Take the reading while the relay is actually on: during that first second, while cranking, or with the engine running. Backprobe. Do not pierce the wire.

Power test TEST REQUIRED

1. Sensor unplugged or backprobed at the harness. Meter negative on a clean engine ground, or on the battery negative.
2. Probe the harness cavity that mates to one heater terminal. Key on. Watch the meter. Repeat on the other heater cavity.
3. The feed side should show about 10–12 V while the ASD is energized. The control side may show voltage only when the PCM is not pulling it down, or a pulsing value. Do not expect both wires to sit at a hard 12 V.
4. Compare with a heater feed at another sensor on the same truck. Power at the others and none here isolates the branch to Bank 2 Sensor 2.

Control / ground test TEST REQUIRED

The PCM switches the heater ground. It is often a pulse, not a solid ground, so a meter may show a changing voltage instead of 0.00 V. With the engine running and the heater feed proven:

- A control wire that the PCM never pulls, while another sensor's heater control does show activity, points at this wire or at the PCM driver for this sensor.
- A control wire that is shorted to ground with the key off and the PCM connector considered can keep the heater on or blow the fuse. Check for a rub-through to the body or the block.
- A control wire shorted to battery voltage will set a circuit fault and can damage a driver. Do not connect it to ground to "make the code go away."

Do not leave a jumper from the control wire to ground. A heater of about 5 Ω draws roughly 2.5 A at 12 V. These heaters are pulsed. A sustained hard ground can overheat the element and the wire. Do not use a jumper as a repair.

Heater ohms	10–12 V feed	PCM control	Action
Open or shorted	Present on the harness	Not the cause	Replace the sensor after oil and connector problems are corrected.
In the 4–10 Ω screen	Missing at this sensor only	—	Repair the feed wire or its connection. Recheck the fuse.
In range	Missing at all sensors	—	Fuse, ASD relay, or ASD output. TEST REQUIRED
In range	Present	Never switches, others do	Repair the control wire. Condemn the PCM only after the wire and the connector test good.
In range	Present	Switches	Look for an intermittent: oil, a spread pin, a wire that opens when the engine rocks. Do not replace the sensor on the code alone.

P0161 · PARTS AND INSTALLATION

When the sensor is actually the part

Replace Bank 2 Sensor 2 when the cold heater is open or shorted, or the sensor-side connector is damaged beyond a terminal repair, and the harness feed has been proven. Fix an oil leak first if oil is reaching this sensor. Guide 09 has the part-number warnings. Short version:

NOT CONFIRMED **NTK 23142** and **Mopar 56029001AB** were checked against published fitment and did not come back as confirmed parts for a 2004 Grand Cherokee 4.0L Bank 2 Sensor 2. Do not buy either number from this sheet. Give a dealer the VIN and the position: rear pipe, after the rear mini-cat. NTK is the usual original supplier of Mopar sensors of this era. **LIKELY** from parts-trade reporting, not a Chrysler letter. This PCM has a poor field reputation with no-name and many Bosch sensors. **POSSIBLE** pattern. Prefer a new NTK or Mopar sensor for the exact bung.

Installation

1. Exhaust fully cool. Raise and support the truck if the bung is reached from below. Wheels chocked, stands rated for the vehicle.
2. Unplug the pigtail from the engine harness. Free the clips. Do not twist the sensor by the wire.
3. Remove it with an oxygen-sensor socket. If it is seized, use penetrating oil on the threads only, not in the connector, and shock the hex. Heat is a last resort on a cold, degreased pipe, with a fire extinguisher at hand and no oil film nearby.
4. A new sensor is already coated. Do not add anti-seize. Extra compound changes the torque and can contaminate the element. Published install torque in the WJ sensor guide is **22 ft-lb (30 N·m)**. **LIKELY** Use a torque wrench. Do not reef it with a breaker bar.
5. Clip the harness back where it was. The wire must not touch the pipe. Connect the plug until the latch clicks.
6. Do not clear codes until Guide 07 says so. After the rest of the repairs, start, confirm there is no fresh oil on the sensor, and rescan. A heater code that returns on the first start is still a circuit problem, not a part that “needs to set.”

Leave the sensor alone when

Ohms are in the screening band, 10–12 V is present while the ASD is on, and the PCM control switches. Spend the time on the connector tension and the harness flex instead. Replacing a good downstream sensor will not fix a high idle, a vacuum leak, or a glowing manifold.

Downstream sensors on this package also help the PCM watch the mini-cats. That function is not what P0161 tests. Signal codes (for example P0157, P0158, P0160) are a different job. Do not retune or delete a sensor to put the light out. This is a California-emissions engine. Repair the circuit.

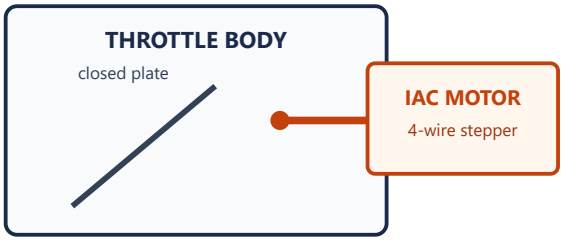
P0505 AND IDLE-AIR CONTROL

IAC operation and P0505

The idle air control motor is a stepper on the throttle body. The throttle plate is closed at idle. Air the engine needs at idle goes through a bypass passage. The PCM moves a pintle in steps and changes how much air gets through.

There is no pintle-position sensor. The PCM counts the steps it has commanded from a reference. It does not "see" where the tip actually is. That is why a pintle moved by hand is a problem even when the coils still measure correctly.

Cold idle is supposed to be higher. A/C, the cooling fan, and an automatic transmission in Drive are loads. The PCM opens the bypass to hold RPM. Judge the motor only when the engine is warm and you have written down the loads.



DO NOT PUSH OR PULL THE PINTLE

Not by hand. Not with a pick. Not "to seat it."

The pintle on this truck was moved by hand STATED

Treat that motor as damaged. Pushing or pulling the pintle can strip the internal screw, bend the tip, or leave the bypass parked where the PCM's step count is wrong. The result can be a high idle, a stall, a hunt, or a circuit fault if the mechanism is jammed. Those are POSSIBLE outcomes, not a list of things already measured. An ohms test cannot see a stripped screw. Replace it with a **new** motor. Do not install a used one. Do not touch the pintle on the new one, in the box or on the truck.

What P0505 is, and is not

CONFIRMED Generic title: idle control system malfunction. On this vehicle family the code is often an **electrical** monitor: a short to ground or to battery on an IAC driver circuit while the motor is commanded. LIKELY Read the scan tool's manufacturer description and save the freeze frame. A high idle alone may set a different monitor, often described as P1294, target idle not reached. TEST REQUIRED P0505 does not, by itself, prove a vacuum leak. Check the circuit and the leaks. Both can be present.

P0505 · VACUUM LEAK VERSUS THE IAC

High idle: leak, or the IAC?

Clue	Points more toward a vacuum leak or a plate that is not shut	Points more toward the IAC
RPM	Stays high even when a scan tool shows the IAC driven to its closed, low-air stop. The exact step count is calibration data. Do not invent a number. TEST REQUIRED	RPM follows IAC command, or the idle is wrong and the pintle was pushed, or the coils measure open or shorted.
Unplugging the IAC	If the pintle is already closed and RPM stays very high, extra air is coming from somewhere else.	If unplugging a motor that was held open drops the idle sharply, that motor was the air path. Freezing a bad position is not a fix.
Sound and smoke	Hiss at an elbow, the booster hose, the intake gasket, or smoke exiting one of those joints.	Smoke only at the IAC base means the O-ring or the mounting face, which is still an air leak at the IAC.
Fuel trims	On this MAP engine, trims may stay near normal because the PCM sees the higher MAP and adds fuel. Trims that go very positive only at idle still deserve a leak and exhaust-leak check.	Trims do not test the stepper windings. Use ohms and the step command for that.
History on this truck	Brittle elbows and a cracked CCV hose are common on this engine. LIKELY pattern. Confirm with smoke.	The pintle was moved. That history is enough to replace the motor even if ohms look fine.

Do the leak check before the idle relearn. A relearn on a leaking engine teaches the PCM the leak. Guide 04. Fix what smoke and the visual inspection find. Then install the new IAC. Then relearn.

IAC resistance TEST REQUIRED

Two coils, four pins. Measure one coil across one pair and the other coil across the other pair. On a straight 4-pin row the usual pairs are the two outer pins and the two inner pins. Several 4.0L procedures cluster near **48–60 Ω per coil**. LIKELY as a field figure, not a 2004 FSM limit in hand. **Screening window for this job: about 40–80 Ω per coil.** The two coils should not read a few ohms to each other. Neither coil should read to the motor case.

Reading	Call
Both coils inside 40–80 Ω, coils isolated from each other	Windings are electrically plausible. If the pintle was pushed, replace the motor anyway. The screw can be ruined with good ohms.
OL on either coil	Open winding. Replace. Do not try to “free” it by pushing the pintle.
Near 0 Ω on a coil, or a few ohms between the two coils	Shorted winding. Replace. Inspect the harness for a short that may have caused it, or that it may have caused.

Key-on voltage at the IAC connector is a PCM driver output, and it changes as the drivers step. Two wires may sit near battery voltage and two near ground, and then swap. A single frozen reading is not a full test. Use it only as a “is anything there” check, then go back to ohms on the motor and a wiggle test of the connector. Wire colors are not listed here on purpose.

P0505 · THROTTLE BODY AND INSTALLATION

Clean the bore. Install a new IAC.

New motor only. Not a junkyard motor, not a motor whose pintle was pushed back in the box, not the original after the pintle was moved. Catalog candidate: Mopar **4874373AB**, listed for Grand Cherokee idle-speed control, 1998–2004. CATALOG LISTED Verify by VIN. This PCM has a reported intolerance for poor aftermarket IACs. POSSIBLE Prefer the new Mopar motor.

Throttle-body inspection and cleaning

1. Key off. Remove the air duct at the throttle body so you can see the plate. Disconnect the IAC connector before any spray.
2. Look for carbon on the plate edge and a plate that does not sit closed. Look at the cable and the cam. The plate must return fully when you open it by the cam and let go. A cable that holds the plate open makes a high idle no IAC can correct.
3. Hold the plate open with the cam. Spray throttle-body cleaner on a shop towel and on the bore. Do not flood the TPS or the IAC connector. Do not pry the plate with a screwdriver.
4. With the IAC removed, clean the bypass passage in the throttle body. Do not spray out a new IAC and do not push its pintle to clean behind the tip. The old motor is not being saved if the pintle was forced.
5. Wipe the bore dry enough that solvent will not be ingested on the first start. Confirm the plate closes and the cable returns.

Installation

1. Negative battery cable off and secured, so the PCM is not stepping a motor you are handling. This also starts the clock for the relearn on the next page.
2. Two screws hold the motor. They are commonly a T20 Torx. Reported torque is **7 N·m (60 in-lb)**. LIKELY Do not overtighten into the aluminum.
3. The O-ring must be present, clean, and not twisted. A missing ring is a vacuum leak at the IAC base. Do not oil it. A dry ring seats.
4. Set the motor straight into the bore. If it does not sit flush, the ring is pinched or the pintle is against the passage. **Do not push the pintle to make it fit.** Remove it and look. The new pintle stays where the factory left it.
5. Snug the screws to the torque above. Connect the plug until the latch clicks. Reinstall the air duct. The duct clamps and the CCV hose at the duct are themselves leak points. Seat them.

After the battery is reconnected, turn the key on and wait about 10 seconds before you crank. Do not touch the accelerator. That key-on pause is the practical way to let this PCM move the pintle before the first start. It is shop practice for this cable-throttle engine, not a quoted FSM script. LIKELY

P0505 · PCM RESET AND IDLE RELEARN

PCM reset and idle relearn

Do this only after leaks are fixed, the throttle plate closes, and the new IAC is installed with the pintle untouched. Resetting first teaches the computer the fault. A scan tool that can clear Chrysler adaptive fuel and idle memory is the cleaner method if you have one. The battery method below is the practical standby.

This is not an electronic-throttle relearn. There is no accelerator-pedal procedure to perform. Foot off the pedal the entire time, except the brake when you select Drive.

1. Negative battery cable disconnected and tied so it cannot touch the post. Wait **15 minutes** minimum. Some radios ask for an anti-theft code afterward. Have that code before you start if this truck uses one.
2. Reconnect the battery. Key on, engine off, about 10 seconds. Do not touch the pedal. Then start.
3. Park, parking brake set, wheels chocked. A/C off, rear defrost off, blower off. Let it idle with no throttle input until the engine is fully warm. That can take 10 to 15 minutes. Do not blip the throttle to “help it.”
4. When warm, idle about two minutes with the A/C on, then turn the A/C off and let it settle.
5. If the brakes hold and the shifter is safe: foot on the brake, idle in Drive about two minutes with loads off, then briefly with the A/C on. Return to Park. Skip this step if the vehicle is not safe to put in gear.
6. Shut off. Restart. Read warm idle in Park, loads off. Screening band **700–800 rpm**. Factory software reference about **744 rpm** fully warm on the TSB calibration. Record RPM, coolant temperature, and gear on Guide 01’s test sheet.

How to read the result

Warm idle after the relearn	Meaning
About 700–800 rpm in Park, loads off, and it recovers after the A/C comes on	Idle control is plausible. **SCREEN MET** Rescan. A pending code still needs the electrical tests.
Still above about 1,000 rpm, no load	Extra air or a motor the PCM cannot close. Go back to smoke, the plate, the cable, and the IAC O-ring. Do not keep resetting.
Hunts more than about 100 rpm, warm, no load	A remaining leak, a dirty plate edge, a poor IAC, or a vacuum hose collapsing. **TEST REQUIRED**
Stalls as it warms, or only with the A/C on	IAC not opening under load, a connector not latched, or an air leak the PCM cannot trim out. Check the plug and command with a scan tool before you buy a second motor.
Idle is normal and P0505 returns immediately	Treat it as the circuit fault: wiring, connector tension, or a PCM driver. A good idle does not clear a shorted driver wire.

If the manifold starts to glow during this idle, shut it off. Guide 05. Do not finish the relearn on an engine that is overheating the exhaust.

VACUUM AND EVAP LEAK GUIDE

How a leak raises idle on this engine

The 4.0L measures load with a MAP sensor. It does not have a mass-airflow sensor. Air that enters through a cracked elbow, a gasket, or a purge valve still changes manifold pressure. The PCM sees that pressure and adds fuel. The engine still speeds up, because the extra air is real and the throttle plate is supposed to be closed. **A vacuum leak is a strong explanation for a high or unstable idle.** LIKELY mechanism. It is not, by itself, proof that the mixture is lean or that the manifold will glow.

High idle — likely

Extra air past a closed plate raises RPM. The IAC steps closed to fight it. If the leak is larger than the IAC's authority, RPM stays high and an idle code can set. Fix the leak before any idle relearn. A relearn on a leaking engine memorizes the leak.

Lean mixture — possible, not automatic

Because the MAP sensor sees the leak, fuel is added with it. The mixture can stay near normal while the idle is wrong. A lean result is still POSSIBLE if the leak, a weak fuel supply, or an injector outruns that correction. Prove it with upstream O2 voltage and fuel trims at idle and at 2,500 rpm. TEST REQUIRED

What a leak can contribute to a glowing manifold

If the added air is not fully matched with fuel, combustion temperature rises and exhaust temperature rises with it. Iron that gets hot enough glows orange or red. That path is physically real. On this truck it is one POSSIBLE path. A misfire that burns fuel in the pipe, retarded timing, and a restricted catalyst can glow the same manifold. Guide 05. Do not close the case at "we found a split hose" if the manifold has been glowing. Confirm the mixture and look for a misfire before you drive it.

Do not spray a flammable leak detector on this truck yet

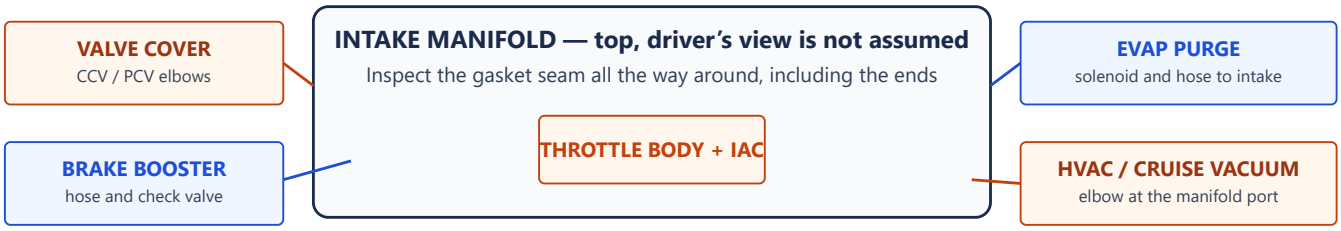
Carb cleaner, brake cleaner, and propane are fire hazards. This vehicle has an oil-on-exhaust concern and a glowing-manifold concern. Those sprays are not the test. Use your eyes, a vacuum gauge, scan data, and a smoke machine. A fire extinguisher stays within reach anyway.

EVAP note: a purge valve that sticks open can act as a vacuum leak and can pull fuel vapor or canister air into the intake. The idle may rise and the trims may swing rich or lean depending on what is in the canister. That is POSSIBLE. Test the valve. Do not replace the canister as a guess.

VACUUM AND EVAP · WHERE TO LOOK

Brittle hoses and elbows

These are common crack points on a 20-year-old 4.0L. **LIKELY** places to inspect. A place on this list is not a leak until you see the crack, hear the hiss, or watch smoke come out of it. Engine cold for the hands-on inspection.



Place	What age does to it	What you should see if it is the leak
CCV / PCV elbows at the valve cover and at the intake or air duct	Hard, split, or swollen from oil vapor. The short rear hose and the front elbow are repeat failures. **LIKELY**	A crack you can open with a finger, or smoke at the fitting. Dorman 47128 is only the short hose set. Compare it. Guide 09.
Hard plastic vacuum lines and molded elbows	They snap instead of bending. Tees crumble.	A white stress crack, or a line that is loose on a port.
Brake-booster hose and check valve	Soft spots, a split at the nipple, a check valve that rattles apart.	A hiss at the hose, a hard pedal, or RPM that falls when that hose is pinched with proper pliers. Pinch only if the hose will not crack.
Intake-manifold gasket	Common on high-mileage 4.0L engines, especially at the ends and lower corners. **LIKELY** pattern.	Smoke at the head-to-manifold seam, or a hiss you can move along the flange with a hose to your ear.
Throttle-body gasket and IAC O-ring	Left off, pinched, or flattened during an IAC job.	Smoke at that joint. Directly relevant if the IAC was recently handled.
EVAP purge hose and solenoid ports	Cracked ports, a hose pulled off, a valve stuck open.	Smoke out of the purge line with the valve closed, or idle that changes when the purge connector is unplugged. **TEST REQUIRED**
HVAC and cruise elbow at the intake	A 2004 factory action covers a narrow build window, 31 March–6 April 2004, where an under-cured elbow can collapse and block HVAC or cruise vacuum. Part in that letter: 55115901AG, elbow and check valve only. Outside that VIN window the same elbow still gets hard and can crack. **POSSIBLE**	A collapsed elbow starves HVAC and cruise. It does not raise idle. A cracked one can. Tell them apart.

VACUUM AND EVAP · SAFE TESTS

Safe leak tests, in order

1. **Cold visual and flex test.** With the engine off and cool, bend every elbow and short hose. Replace anything cracked, oil-soft, or so hard it has gone white at a fitting. Tug each connection. A hose that stays on only by habit will leak when the engine rocks.
2. **Listen, after the fire risk is down.** Only if the manifold is not glowing and oil has been cleaned off the exhaust. A length of hose used as a stethoscope along the intake seam, the booster hose, and the purge line. A sharp hiss that stops when you press a fitting is a find.
3. **Smoke test, engine off.** This is the preferred test. Introduce smoke into a manifold vacuum port. Block the intake inlet lightly so the smoke stays in the manifold. Watch the booster hose, the EVAP line, the intake gasket seam, the IAC base, the throttle-body gasket, and every CCV fitting. A smoke machine uses an inert or low-temperature vapor. It is the right tool while oil and heat are part of the story.
4. **Vacuum gauge, warm idle, only after a safe start is allowed.** Near this elevation a healthy 4.0L typically shows about **17–21 inHg**, steady. LIKELY screen, not a factory limit. A needle that flickers can be a leak or a misfire. A low steady reading can be a leak, late timing, low compression, or a restricted exhaust. Record the number. Do not condemn a gasket on the gauge alone.
5. **Scan data.** MAP, RPM, short-term and long-term fuel trims at idle and at about 2,500 rpm, upstream O2 voltage. On speed-density, a leak often shows up as high RPM with MAP higher than a normal idle and trims that are not dramatically positive. Trims that are largely positive only at idle, and come back at higher RPM, still suggest a small leak or an exhaust leak ahead of an upstream sensor. Write the numbers on the test record. TEST REQUIRED
6. **Pinch test, selective.** Pinch one suspect hose at a time with hose pliers while watching RPM. A drop in RPM means that hose was admitting air. Do not pinch a hard plastic line. Do not pinch the booster hose if the pedal is your only brake and the engine is the only source of assist you are relying on while the truck is not secured. Wheels chocked, Park, parking brake set.

Not approved on this job

Spraying carburetor cleaner or brake cleaner around the intake. Introducing propane under the hood. Those methods can find a leak on a clean, cool engine. This engine has an oil film and a possible extreme exhaust temperature. Skip them. If someone already did it, do not repeat it.

A found leak is still one finding

Repair it, then install the new IAC if the pintle was moved, clean the throttle body, and do the relearn. Verify idle. If the manifold glowed before the repair, watch it on the first idle from outside the wheel well, extinguisher ready, and shut down if color returns to the iron. Guide 05.

VACUUM AND EVAP · PCV, CCV, AND EVAP PARTS

PCV, CCV, and EVAP

The 4.0L cranksase ventilation is a hose-and-elbow system, not a single round PCV valve hidden under a rubber grommet the way older engines were. Oil vapor moves from the valve cover to the intake. The elbows harden, split, and then become both a vacuum leak and an oil leak onto the manifold. That ties Guide 04 to Guide 06.

CCV / PCV — inspect all of it

- Front elbow at the valve cover.
- Rear hose and grommet at the valve cover. This is the piece people find split.
- The hose from the cover to the intake or to the air duct.
- Any small hose that also tees toward the MAP sensor. Dorman 47128 is catalog-listed as that short dual-hose set, about 2.56 in, two hoses, OE cross 4854265. **CATALOG LISTED** It is not the whole system.
- Grommets that are loose in the cover. A new hose on a loose grommet still leaks.

EVAP — inspect, don't shotgun

- Purge solenoid on the engine. Cracked plastic ports are common once the hose has been yanked.
- Hose from the solenoid to the intake port. Hard, split, or off the nipple.
- Hose from the solenoid back toward the canister. A break there can set EVAP codes and may not change idle. Different fault.
- A stuck-open purge valve is an idle-air path. Unplug it only as a comparison, with the scan tool recording RPM. If idle falls and stays down, test or replace the valve. **TEST REQUIRED**
- Do not replace the charcoal canister, the leak-detection pump, or the fuel-tank seals because of P0505. Those parts belong to an EVAP-code diagnosis.

Repair rules

1. Replace the failed piece with hose rated for fuel and oil vapor, the right inside diameter, and a length that does not kink. A hardware-store vacuum hose of the wrong size is another leak.
2. Match the part in your hand to the catalog picture before you buy. Guide 09. Verify by VIN. The number 53030006 did not check out as a Mopar elbow for this Jeep. **NOT CONFIRMED**
3. Do not block a PCV port to "cure" a high idle. The crankcase then pressurizes, oil comes out of the next seal, and that oil lands on the exhaust.
4. After the hoses are sealed, smoke-test once more. Then, and only then, install the new IAC if it is not already on, and do the relearn in Guide 03.

MAP sensor. On this engine it is typically mounted in the intake, not on a long hose. Look for a cracked port or a sensor that is loose in the bore. A vacuum hose you "should" find may not exist on this manifold. Do not add one. If the sensor or its seal is leaking, smoke will show it. A MAP sensor that reads wrongly can also destabilize idle. Compare key-on MAP (it should read barometric, roughly 98–100 kPa near this elevation, before the engine starts) with idle MAP. **TEST REQUIRED** A failed MAP is **POSSIBLE** and is not the leading theory on a truck whose IAC pintle was pushed.

GLOWING EXHAUST MANIFOLD — EMERGENCY

STOP DRIVING · SAFETY CONDITION · DO NOT IDLE IT TO SEE

The manifold is glowing

Bright orange or red exhaust iron means the pipe is hot enough to crack the casting, burn valves, melt a catalyst, destroy oxygen sensors, and ignite oil. That is a stop. Park it outdoors, away from the building and dry grass if anything is smoking. Engine off. Transmission in Park. Parking brake set.

Do this

- Get people away from the wheel well.
- ABC or BC extinguisher within reach. If you see flame, get clear and call the fire department.
- Let it cool completely. An hour is a minimum, not a guarantee. The cast manifold holds heat.
- Do not open the radiator cap. Do not spray water on red iron. Thermal shock cracks it.
- Do not drive it to a shop. Tow it if it has to move.

Glow versus old stain

An active glow is bright orange or red while the engine is running or has just been shut off. Brown, blue, or purple color on a cold manifold is heat stain from some earlier event. Stain is a clue. POSSIBLE past overheat. It is not the same as watching the iron glow. If you saw it glow, that observation stands even after the color fades.

Possible causes — none of these is confirmed

Cause	Why the iron gets hot	Label
Lean combustion	More air than the fuel that was delivered. Combustion temperature and exhaust temperature rise.	POSSIBLE TEST REQUIRED
Misfire, fuel burning in the exhaust	A cylinder that does not fire dumps raw fuel into the manifold. The fuel lights there. This is a classic way to make a manifold glow, and it can happen with a rich cylinder, a dead coil, or a fouled plug.	POSSIBLE TEST REQUIRED
Retarded ignition timing	Late spark pushes burning gas into the exhaust. The 2004 4.0L has a coil rail and no distributor. Timing is not “set” with a wrench. A crank or cam signal problem is the sort of fault that belongs here.	POSSIBLE
Restricted catalyst	A melted mini-cat or main cat holds heat in the manifold. Often the result of a misfire or of oil burning, not the first event. Low power and vacuum that falls as RPM rises are clues.	POSSIBLE TEST REQUIRED

More than one of these can be true. A vacuum leak can start the idle problem. A misfire can be what actually glows the iron. Test both. Do not cut a catalyst out as a test. This is a California-emissions engine. Repair it.

GLOWING MANIFOLD · LEAN PATH AND DAMAGE

Vacuum leak, lean mixture, extreme heat

Working theory, not a closed case. **POSSIBLE**

A vacuum leak adds air. On this MAP engine the idle rises because that air is real. If the injectors and the MAP calculation do not add enough fuel to match it, the mixture goes lean. Lean combustion runs hotter. Exhaust temperature can climb high enough to make the cast manifold glow. That chain is physically possible. It is not established on this truck until fuel trims and upstream oxygen-sensor voltage show a lean condition, and until a misfire and a restricted catalyst have been considered.

What continued running damages

- The manifold itself: cracks at the collector and at the mounting flange.
- Exhaust valves and seats.
- The front and rear mini-catalysts, and the main catalyst. A melted cat then becomes the restriction that keeps the next run just as hot.
- Oxygen sensors. A sensor cooked by a glow, or soaked in the oil that the heat ignited, will not come back.
- Under-hood wiring, the starter, and any oil film. Oil plus a glowing manifold is how this becomes a fire. Guide 06.

After it has cooled — the only tests that are in bounds

- Visual: cracks in the manifold, a cherry-to-gray fresh scale, oil washed onto the iron, a sensor wire melted to the pipe.
- Do not start it for a “short check” until oil has been cleaned off the exhaust and a fire extinguisher is out.
- When a start is allowed: watch from the side, not over the manifold. Idle only. If any orange returns, key off. Tow it.
- Read misfire counters if the scan tool has them. Read upstream O2 and fuel trims. A single glowing event with a dead cylinder is a misfire case until proven otherwise.
- Catalyst check, only on a later and cooler run: infrared temperature ahead of and behind each cat, or a pressure gauge in front of the cat. Do this after the idle and the oil leak are under control, not on the day the manifold was red.

Return-to-service rule. The manifold stayed dark on a full warm idle, oil is not reaching the pipe, warm idle is in the 700–800 rpm screen, and a rescan shows no new misfire or fuel-control fault. If any one of those is missing, it is not finished. If the glow returns, stop immediately. Do not reset anything and try again.

OIL LEAK AND FIRE HAZARD

STOP IF OIL IS REACHING HOT EXHAUST OR AN OXYGEN SENSOR

Oil on the manifold is a fire hazard

Engine oil on hot cast iron, on a downpipe, or on an oxygen sensor can ignite. A manifold that has been glowing makes that ignition likely, not theoretical. This problem is separate from the idle code and separate from P0161. It can destroy the engine and the vehicle on its own.

If you see smoke or flame from the oil

Engine off. Everyone out and away. If there is flame, do not open the hood into a fireball if the fire is already rolling out of it. Call the fire department. Use an extinguisher only if you can do it from a safe direction and you have a clear exit. Do not drive the truck away from the fire.

After it is cool — find the fresh leak

1. Confirm the iron is cold enough to hold a hand on. A manifold that glowed needs a long cool-down. Do not degrease a hot pipe. Solvent on hot iron is another fire.
2. Degrease the passenger side, where this engine's manifold and oil-filter adapter live. Clean the valve-cover rail, the filter area, the block side, the sensor connectors, and the top of the pipe. Brake cleaner or a purpose-made degreaser, cold engine, ventilation, no smoking, rags out of the bay and into a metal can. Do not soak electrical connectors. Wipe them.
3. Start only when the pipe is clean and has not been glowing. Idle. Watch for the first wet line of new oil. That line is the source. Old black film lies about direction. Fresh oil does not. UV dye is the right tool if the leak is a seep rather than a run.
4. Repair that source before any oxygen sensor that was sitting in the oil is replaced. A new sensor in the same spray fails again, and the fire hazard is still there.

P0161 does not diagnose the leak, and the leak does not diagnose P0161. Oil in the connector can corrode the heater terminals and cause the code. **POSSIBLE** Oil on the sensor body more often kills the signal than the heater. Test the heater circuit in Guide 02 after the area is clean and dry. Replace the sensor if the test says the heater or the connector is finished. Leave it alone if the circuit tests good.

OIL LEAK · COMMON 4.0L SOURCES

Where a 4.0L usually leaks

Every item below is a common source on this engine. LIKELY as a place to look. None is confirmed until the fresh oil is traced to it. Order the gasket or seal by VIN after you know which one. No part numbers are printed here, so the wrong gasket does not get bought from this page.

Source	Where the fresh oil shows	Trap to avoid
Valve-cover gasket	Passenger-side rail and the corners. Oil runs down the cover and onto the manifold. Often the leak that wets the exhaust highest up. Very common. LIKELY	The cover bolts loosen. Check torque and the bolt seals before you condemn only the gasket. A CCV elbow leaking oil is next to this and is a different part.
Oil-filter adapter O-rings	The adapter between the filter and the block. Oil runs down the block, onto the exhaust, and toward the starter. A classic 4.0L leak. LIKELY	This is often miscalled a rear main. If the oil is at the filter housing, do the adapter. The job is the O-rings, and the surface has to be clean.
Oil-pressure sending unit	The sender threads into the block. A weep runs down from the hex and can look like the adapter or the pan.	Watch whether the wet starts at the sender threads. Replace the sender if it is the source. Do not add thread tape that can enter the oil passage. Follow the sender's seal design.
Oil-pan gasket	Corners and the side rails. Drips lower than the manifold. Road speed can blow it back onto the pipe.	Clean first. A pan leak and an adapter leak can both be present. The higher wet point is the one hitting the manifold at idle.
Rear main seal	Between the engine and the transmission, seen at the inspection cover or as oil thrown around the bellhousing.	Do not replace this seal until the adapter, the pan rail, and the sender have been ruled out with a clean-and-watch. It is the expensive leak, and it is not the most common one on this engine.

Repair the leak before the sensor

When oil has been on an oxygen sensor or in its connector, the leak repair comes first. Then dry the connector, then run the Guide 02 tests. Replace the sensor only if those tests fail, or if the sensor shell and the connector are physically contaminated and cannot be made reliable. A heater that ohms correctly and has power and control does not get replaced "because there was oil nearby."

Prove the repair

After the gasket or seal is done, degrease again. Idle. Look. Then a short drive only if the manifold has stayed dark and no new oil is tracking toward the pipe. Check again when you park. A leak that only appears at speed still needs the UV dye. Recheck the sensor connector for fresh oil before you clear P0161.

Passenger-side note: the 4.0L exhaust manifold and the oil-filter adapter are on the passenger side of this chassis. Confirm that on the truck. A leak on the other side can still blow oil onto a pipe, but the first look is the manifold side.

STEP-BY-STEP REPAIR PRIORITY

Do the work in this order

Skipping ahead replaces parts on a truck that may still be a fire risk, and it teaches the PCM a leak. Each line is the action. The note under it is why, and how sure we are.

1. **STOP DRIVING** if the manifold is glowing or oil is reaching hot exhaust. STOP
   Active orange or red iron, or oil on the pipe or a sensor, is a safety stop. Guides 05 and 06. Do not idle it to confirm.

2. **Let the vehicle cool completely.**
   An hour is a minimum after a glow. Do not spray water on the manifold. Do not degrease a hot pipe. Solvent on hot iron is a fire.

3. **Find and fix the vacuum leak.** TEST REQUIRED
   Smoke, visual, and the brittle CCV, EVAP, booster, and intake joints in Guide 04. A leak is a LIKELY cause of high idle on this MAP engine. It is only a POSSIBLE cause of a lean glow. Fix what you find. Do not relearn yet.

4. **Replace the damaged IAC with a NEW correct unit.** CATALOG LISTED
   The pintle was moved. Treat the motor as damaged even if the coils ohm in range. Mopar 4874373AB is the catalog candidate for a 1998–2004 Grand Cherokee. Verify by VIN. No used motors.

5. **Do NOT move the new IAC pintle.** RULE
   Not in the box, not to “seat” it, not to clean behind the tip. The PCM counts steps. A pushed tip makes the count a lie. If the new one does not sit flush, look for a pinched O-ring. Do not force the pintle.

6. **Clean the throttle body.**
   Plate edge and bore, key off. The plate must close and the cable must return. A cable that holds the plate open will not be fixed by any IAC. Do not soak the TPS.

7. **Reset the PCM and perform the idle relearn.**
   Only after steps 3 through 6. Battery negative off 15 minutes, or a scan tool that clears adaptive memory. Key on 10 seconds, no throttle, then the warm idle procedure in Guide 03. This is a cable-throttle engine. There is no pedal relearn.

REPAIR PRIORITY · FINISH AND PROVE

8 Verify warm idle. TEST REQUIRED

Fully warm, Park, loads off. Screening band about 700–800 rpm. TSB reference about 744 rpm on the cited 2004 calibration. Record RPM and coolant temperature. Above about 1,000 rpm means go back to the leak, the plate, and the IAC O-ring. Do not reset again and hope.

9 Locate and repair the oil leak. TEST REQUIRED

Degrease cold. Watch fresh oil. Common 4.0L sources: valve cover, filter-adapter O-rings, oil-pressure sender, pan gasket, rear main. LIKELY places. The fresh track decides. Do not replace a rear main until the adapter and the pan are ruled out.

10 Recheck and repair the P0161 heater circuit. TEST REQUIRED

Separate fault. Bank 2 Sensor 2, rear downstream, after the rear mini-cat. Connector, fuse, cold heater ohms (screen 4–10 Ω), 10–12 V while the ASD is actually on, then PCM control. Replace the sensor only if the element or the sensor connector failed, and only after oil is no longer reaching it.

11 Clear codes.

Only after the test record is filled and the repair those tests support is done. Freeze frame is gone once you clear. Photograph the scan tool first.

12 Start, monitor, and rescan.

Idle until warm. Watch the manifold from the side. Watch for a new oil track. Read RPM. Rescan. A code that returns on this start is still a real fault.

13 If the manifold glows again, STOP immediately. STOP

Key off. Cool it. Tow it. Do not repeat the relearn. Go back to Guide 05: lean condition, misfire, timing signal, and a restricted catalyst are all still open until tested.

Finished only when all of these are true

The manifold stays dark. No fresh oil reaches the pipe or a sensor. Warm idle is in the 700–800 rpm screen in Park with loads off. P0505 and P0161 stay off after the drive cycle you can safely complete. Anything less is an unfinished test, not a comeback mystery.

The order above is the repair priority for this truck's stated history: a pintle that was moved, a high-idle / P0505 concern, a possible lean-heat path, a separate oil leak, and a separate P0161. It is not a claim that every arrow in that history has been measured.

DIAGNOSTIC FLOWCHART · IDLE AND P0505

P0505 decision tree

P0505 stored? Save freeze frame before anyone clears it.

▼

Check idle RPM — warm, Park, loads off
Screen 700–800 rpm. TSB reference about 744 rpm. TEST REQUIRED

▼

High or unstable?

YES
Check vacuum leaks. Guide 04. Smoke and visual before any parts. A leak is a likely idle cause. It does not prove a lean mixture.

▼

Inspect the IAC
Both coils. Screen 40–80 Ω. Connector latched. O-ring present.

▼

Was the pintle moved by hand?
YES on this truck — stated. Treat the motor as damaged even if ohms are in range.

▼

Replace with a NEW correct IAC
Do not touch the new pintle. No used motor. Verify 4874373AB by VIN.

▼

Clean the throttle body
Plate must close. Cable must return.

▼

Reset PCM / idle relearn
After the leak is fixed. Guide 03. Foot off the pedal.

▼

Verify warm idle
700–800 rpm screen. Rescan. If still high, go back to smoke. Do not buy a second motor yet.

NO — idle is already in the screen
P0505 is then the electrical monitor until proven otherwise. A normal idle does not erase a shorted driver wire.

Electrical branch

Inspect the connector and wiggle the harness with the code being watched. Check both IAC coils. A short to ground or to battery on a driver wire is the Chrysler-family description of this code. LIKELY TEST REQUIRED

Compare all four circuits. Condemn the PCM only after the motor and the wires test good. Technician reports say a poor aftermarket IAC will bring this code back. POSSIBLE

Also look for P1294, target idle not reached, if the tool lists it. High idle with that code still sends you up the YES column.

Stop the tree

If the manifold glows at any box, key off. Leave this page. Guide 05. If oil is on the exhaust, Guide 06 before the road test that would “verify” the idle.

Unplugging the IAC freezes the pintle where it sits. It is a comparison, not a repair. A motor that was pushed does not get plugged back in for a second chance.

DIAGNOSTIC FLOWCHART · P0161 HEATER CIRCUIT

P0161 decision tree

Bank 2 Sensor 2. Rear downstream sensor, after the rear mini-cat, before the main cat. **LIKELY** Confirm the bung. This tree does not diagnose idle and it does not diagnose a glow.

P0161 stored?

▼

Inspect connector and wiring
Oil, corrosion, spread pins, melted insulation, harness touching the pipe. Cold engine.

▼

Check the fuse
ASD fuse from the under-hood lid. Measure it. Do not guess the cavity.

▼

Check heater resistance — sensor cold
Screen about 4–10 Ω on the sensor side. OL = open heater. Near 0 Ω = shorted heater.

▼

Check 12 V supply
About 10–12 V while the ASD is actually on. A steady key-on 0 V can be the relay already dropped out.

▼

Check PCM ground / control
The other heater wire. Often pulsed, not a hard ground. Compare with another sensor.

▼

Repair the wiring, fuse, or connector
OR replace the sensor if the heater element is open or shorted.

Oil gate

If oil is in this connector or on this sensor, fix the leak and dry the plug before a new sensor goes in. Otherwise you will buy this part twice. Guide 06.

Ohms	Power	Control	Do this
Open or short	Harness has feed	—	Sensor. After the oil gate.
4–10 Ω	Missing here only	—	Repair this feed wire.
4–10 Ω	Missing everywhere	—	Fuse or ASD relay.
4–10 Ω	10–12 V	Dead, others work	Control wire, then PCM.
4–10 Ω	10–12 V	Pulls / pulses	Intermittent connector or wire. Do not replace yet.

Not this code

A heater that tests good will not fix a high idle, a vacuum leak, or a glowing manifold. Signal codes are a different test. Do not swap sensors between banks. The pigtail lengths are not the same.

Parts gate NOT CONFIRMED

Do not order NTK 23142 or Mopar 56029001AB from this sheet. Neither checked out as the 2004 Grand Cherokee 4.0L Bank 2 Sensor 2. Guide 09. VIN and position at the parts counter.

Do not hard-ground the control wire and leave it. A heater around 5 Ω draws about 2.5 A. The PCM pulses it for a reason. No jumper repairs.

PARTS AND REFERENCE SHEET

VERIFY EVERY PART NUMBER BY VIN BEFORE YOU BUY

A four-sensor California 4.0L and a two-sensor federal 4.0L do not share oxygen-sensor part numbers or harness lengths. Count the sensors first. The position is part of the order: front or rear, before or after the mini-cat.

Oxygen sensors — 2004 4.0L numbers that were actually published

Source: JeepSpecs WJ oxygen-sensor guide, a compiled reference, not a dealer VIN lookup. The page lists four numbers for the 2004 4.0L and says federal and California are distinguished in the original table by color. That color key did not survive as a reliable position map. Hand all four to the counter with the VIN. Do not buy a set.

Published number	How the guide groups it	Status
56028995AA	2004 4.0L, before catalyst, one of two	CANDIDATE Position not proven here
56028994AA	2004 4.0L, before catalyst, one of two	CANDIDATE Position not proven here
56028997AA	2004 4.0L, after catalyst, one of two	CANDIDATE Position not proven here
05139021AA	2004 4.0L, after catalyst, one of two	CANDIDATE One owner reported this box as the rear downstream, sensor body stamped 56044598AA. Single report. Not a confirmation.

Numbers supplied for this job that did not check out

NTK 23142. Catalog listings found for this number are other Jeeps: 2001–2003 Wrangler / TJ downstream, Bank 1 Sensor 2, and 2002–2003 Liberty Bank 2 Sensor 2. Not a 2004 Grand Cherokee 4.0L listing. NOT CONFIRMED Do not buy it unless the dealer or the NTK catalog for this VIN and this bung returns 23142.

Mopar 56029001AB. No fitment for this vehicle was found. Nearby real numbers (56028994AA, 56029049AA, and the four above) are different parts. NOT CONFIRMED Do not order 56029001AB from this sheet.

If the heater test says the sensor is bad

Order one sensor, for Bank 2 Sensor 2, rear pipe after the rear mini-cat, by VIN. Prefer new Mopar or NTK. NTK supplied many Mopar sensors of this era. LIKELY from the parts trade, not a factory letter. Install torque cited in the WJ guide: 22 ft-lb (30 N·m). Do not add anti-seize. Do not install it into an oil spray. Guide 02 and Guide 06.

PARTS AND REFERENCE · IAC, HOSES, TEST SCREENS

VERIFY ALL PART NUMBERS BY VIN

Item	Number	What the source actually says	Status
IAC motor, new	Mopar 4874373AB	Dealer catalog describes this as the idle-speed control motor for Jeep Grand Cherokee 1998–2004, and Wrangler 1998–2004. Aftermarket crosses such as Standard AC176 appear in listings. This PCM is reported to reject poor aftermarket units.	CATALOG LISTED Verify VIN. Buy new. Do not push the pintle.
PCV / CCV short hose	Dorman 47128 Chrysler 4854265	Dorman lists 47128 for 1997–2004 Grand Cherokee, including 4.0L. Two hoses, about 2.56 in, intake manifold to valve cover and MAP. OE cross 4854265. It is a small hose set, not the entire crankcase hose.	CATALOG LISTED Match it to the piece in your hand.
Vacuum elbow example	53030006	No Mopar vacuum-elbow application for this Jeep was found. A nearby number, 53030008, is a vacuum tee listed for a PT Cruiser, which is the wrong vehicle. Marketplace ads cite other elbow numbers. None of that is a confirmation.	NOT CONFIRMED Do not buy this number from this sheet.
HVAC / cruise vacuum line	55115901AG	Factory rapid-response 04-008, only for 2004 WJ 4.0L built 31 March–6 April 2004. Replaces a collapsing elbow and check valve at the intake, not the whole harness. That failure blocks HVAC and cruise vacuum. It does not cause a high idle.	ONLY IF VIN IS IN THAT WINDOW
Valve cover, filter-adapter O-rings, oil-pressure sender, pan gasket, rear main	Order by VIN after the leak is identified	No number is printed here. Buying a rear main for an adapter leak is the usual expensive miss on this engine.	TEST REQUIRED

Test screens used in this package

Test	Screen	How hard this number is
Warm idle, Park, loads off	About 700–800 rpm	Job screen. TSB 18-008-04 states about 744 rpm fully warm for domestic 2004 4.0L trucks built before 6 March 2004. CONFIRMED as the bulletin text. Not proof this VIN has that software.
IAC, each coil, isolated from the other	About 40–80 Ω. Field notes often 48–60 Ω.	LIKELY screen. Not a 2004 FSM page in hand. Good ohms do not save a pushed pintle.
O2 heater, sensor cold	About 4–10 Ω. Earlier 4.0L tests cite about 4–7 Ω.	LIKELY screen. Open or near zero is a failed heater. In-range means keep testing power and control.
O2 heater power	About 10–12 V while the feed is energized	Earlier 4.0L key-on tests use 10–12 V. ASD may drop out a second after key-on. TEST REQUIRED
Idle vacuum, near 700 ft	About 17–21 inHg, steady	LIKELY screen. Low or flickering has several causes.
IAC screws	7 N·m (60 in-lb) reported	LIKELY Do not strip the throttle body.
O2 sensor	22 ft-lb (30 N·m) in the WJ guide	LIKELY No extra anti-seize on a coated new sensor.

Brand note for this PCM. Oxygen sensors and IAC motors are the two parts this generation is repeatedly reported to be picky about. A new Mopar or NTK part, correct position, verified by VIN, is the purchase. A used IAC, a pushed pintle, or a universal oxygen sensor with cut wires is how the same codes come back.

ONE-PAGE QUICK REFERENCE · WORKING THEORY IS NOT A CONFIRMED DIAGNOSIS

STOP DRIVING IF THE MANIFOLD IS GLOWING OR OIL IS ON THE EXHAUST. COOL COMPLETELY.

Glowing manifold

STOP Bright orange or red iron. Crack, burned valves, melted cats, fire if oil is present. Do not spray water. Do not drive it. Tow if it has to move. If it glows again, stop immediately.

Causes still open: lean burn, misfire afterburn, retarded timing, restricted cat. **POSSIBLE** each. **TEST REQUIRED**

Oil / fire

STOP Oil on hot iron or on an O2 sensor can ignite. Separate problem from the idle code. Degrease only when cold. Trace fresh oil.

Common 4.0L sources: valve cover, filter-adapter O-rings, pressure sender, pan, rear main. **LIKELY** places. Repair the leak before replacing an oil-soaked sensor.

IAC — do not touch the pintle

Stepper motor. PCM counts steps and cannot see the tip. This pintle was moved. **STATED** Treat it as damaged. New motor only. Never push, pull, or pry the new pintle.

Candidate: Mopar 4874373AB. **CATALOG LISTED** Verify by VIN. Coils that ohm fine do not save a stripped screw.

Vacuum leak

MAP engine, no MAF. A leak raises idle. **LIKELY** It does not automatically make the mixture lean, because the PCM sees the MAP change and adds fuel. Lean is **POSSIBLE** and must be measured. Smoke test. No carb-cleaner spray while oil or a glow is in play. Fix leaks before the relearn.

P0505

CONFIRMED title: idle control system. Often an IAC circuit short on this family. **LIKELY** A high idle is a separate fact and does not prove the code, or a vacuum leak.

Path: RPM, leaks, IAC ohms, pintle history, new IAC, clean bore, relearn, verify. Guide 03 and Guide 08.

P0161

CONFIRMED Bank 2 Sensor 2 heater circuit. Rear downstream, after the rear mini-cat. **LIKELY** Not a mixture code. Not the cause of a high idle.

Connector, fuse, cold ohms, 10–12 V while ASD is on, PCM control. NTK 23142 and 56029001AB are **NOT CONFIRMED** for this truck.

WARM IDLE

700–800 rpm screen, Park, loads off. TSB about 744 rpm. **TEST REQUIRED**

IAC COILS

About 40–80 Ω each. Often 48–60. Isolated from each other. **LIKELY**

O2 HEATER

About 4–10 Ω cold. Earlier spec about 4–7 Ω. OL = open. **TEST REQUIRED**

HEATER POWER

About 10–12 V while ASD is on. Key-on 0 V can be a false fail.

Repair order

1. Stop if glowing or oil is on hot exhaust.
2. Cool completely. No water on the iron.
3. Find and fix the vacuum leak. Smoke, not spray.
4. New IAC, correct unit, verified by VIN.
5. Do not move the new pintle.
6. Clean the throttle body. Plate must close.
7. PCM reset and idle relearn. Foot off the pedal.
8. Verify warm idle. Record RPM.
9. Find and repair the oil leak. Fresh oil decides.
10. Retest P0161. Repair wire or sensor as the test says.
11. Clear codes only after the tests are written down.
12. Start, watch the manifold and the oil, rescan.
13. If it glows again, stop immediately and tow.

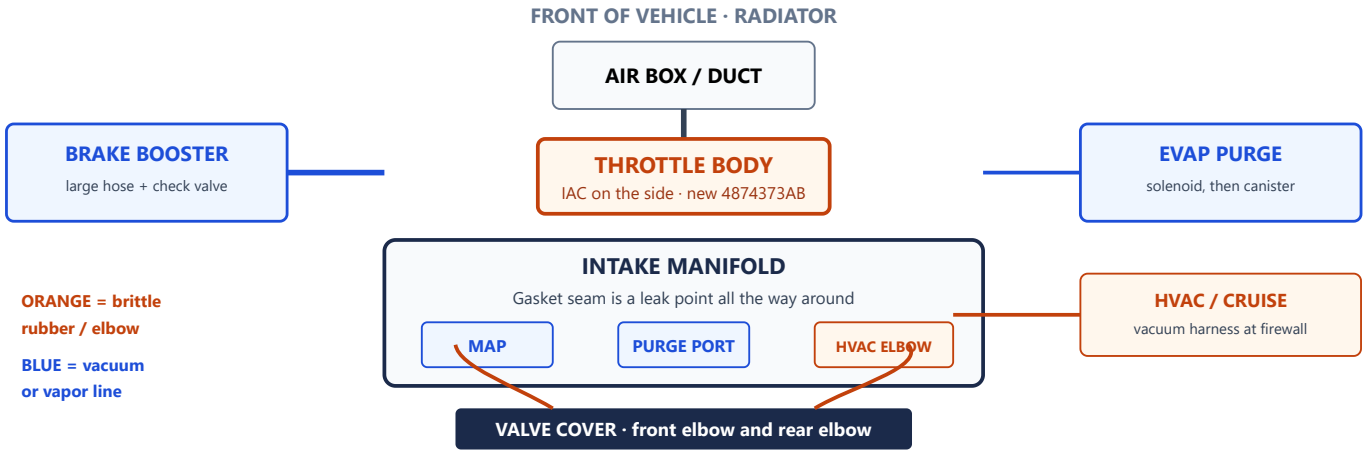
Guides 01–09 have the procedure for each line.

Possible chain, not confirmed: pintle moved → IAC damaged or held open → extra air → high idle → P0505 → mixture may go lean → exhaust may glow. Prove each step. A misfire can glow the manifold without that chain.

Two other problems stay separate: oil leak → fire on hot iron. And P0161 → heater circuit of Bank 2 Sensor 2 → test sensor, power, control, fuse, wiring. Do not trade one repair for another.

HOSE MAP · TYPICAL 4.0L ROUTING · CONFIRM EVERY LINE ON THIS TRUCK

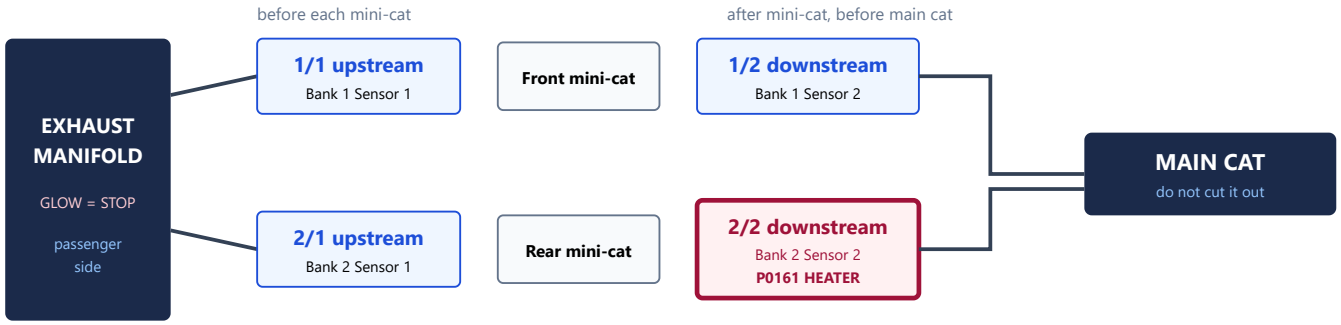
THIS IS A WORKING MAP, NOT A FACTORY VACUUM DIAGRAM · MATCH THE HOSE IN YOUR HAND



Line	Where it runs	Part to compare
CCV / PCV elbows	Valve cover, front and rear, into the intake or the air duct. The rear piece is short and often split. Oil vapor and vacuum both use this path.	Dorman 47128 / 4854265. Two short hoses, about 2.56 in. CATALOG LISTED Not the whole harness.
Brake booster	Large hose from the booster to a manifold port, with a check valve in the line. A split here raises idle and can make the pedal hard.	Match the hose and the valve. No number on this sheet.
EVAP purge	Purge solenoid on the engine, one hose to an intake port, one hose back toward the canister. A stuck-open valve acts like a vacuum leak.	Replace the cracked hose or the solenoid only after the test. Do not buy a canister for P0505.
HVAC / cruise	Small vacuum harness. The elbow and check valve are at the intake. A collapsed elbow kills HVAC and cruise. A cracked elbow can raise idle.	55115901AG only if the VIN is in the 31 Mar–6 Apr 2004 window. NARROW
IAC base	Not a hose. The O-ring under the motor is an air leak if it is missing, twisted, or flattened.	Comes with a new IAC. Do not push the pintle to seat it.

Ports on the drawing are grouped so the lines are readable. They are not a measured left/right layout. Fuel lines, heater hoses, and the power-steering hoses are not on this map. Do not pull those while chasing a vacuum leak.

PASSENGER-SIDE EXHAUST · FOUR SENSORS IF THIS IS THE CALIFORNIA PACKAGE



Oil that can reach this pipe

Trace fresh oil after a cold degrease. **LIKELY** sources, none confirmed until the wet trail starts there: valve-cover rail above the manifold, oil-filter adapter O-rings on this same side, oil-pressure sender, pan rail, rear main at the bellhousing. Fix the leak before a new O2 sensor goes in. No gasket numbers on this sheet. Order by VIN after the source is known.

Sensor purchase rule

Count sensors first. Two means federal. Four means the California layout drawn here. Buy one sensor, for the bung that failed the test, by VIN. Published 2004 4.0L candidates, position not mapped: 56028995AA, 56028994AA, 56028997AA, 05139021AA. **CANDIDATES** NTK 23142 and Mopar 56029001AB are **NOT CONFIRMED**.

Callout	What it is	Buy it when
1/1 and 2/1	Upstream sensors, before the mini-cats. These trim fuel. **LIKELY**	A signal or heater code names that exact sensor, and the circuit test fails the sensor.
1/2	Front downstream, after the front mini-cat. Watches that mini-cat.	Same rule. P0161 is not this sensor.
2/2	Rear downstream. Heater circuit is P0161.	Cold heater is open or shorted, or the sensor-side plug is destroyed, and oil is no longer hitting it.
Mini-cats and main cat	Three catalysts on the California package. **LIKELY**	Only after a temperature or pressure test shows a restriction. Not a first part. Do not delete them.
Manifold	Cast iron on the passenger side. A glow is a stop, not a parts order.	Replace only if it is cracked after it has been cool long enough to inspect. Guide 05.

Harness length and connector key are different at each bung. Do not swap sensors to see what happens. Clip every pigtail back so the wire cannot touch the pipe. Splice a broken wire with an adhesive-lined butt connector, several inches off the iron, staggered, then put the loom back.

PARTS LIST FOR THIS JOB · VERIFY EVERY NUMBER BY VIN

DO NOT HAND THIS PAGE OVER AS A BLIND ORDER · VIN AND THE PART IN YOUR HAND DECIDE

Item	Number	Use it for	Status
IAC motor, new	4874373AB	Pintle was moved, or a coil is open or shorted. Do not push the new pintle.	CATALOG LISTED
Short CCV hoses	Dorman 47128 Chrysler 4854265	The small cracked elbows at the valve cover / intake / MAP. About 2.56 in, two hoses.	CATALOG LISTED
O2, before catalyst	56028995AA 56028994AA	2004 4.0L published list. Which bank is which was not recoverable.	CANDIDATE
O2, after catalyst	56028997AA 05139021AA	Same list. One owner report tied 05139021AA to the rear downstream. Not proof.	CANDIDATE
HVAC vacuum line	55115901AG	Elbow and check valve only. Build dates 31 Mar–6 Apr 2004. Blocks HVAC and cruise. Does not fix a high idle by itself.	VIN WINDOW
Booster hose, purge hose, intake gasket, throttle-body gasket	Match by VIN	Buy the one the smoke test or the crack identifies. Not a set.	TEST REQUIRED
Valve cover, filter-adapter O-rings, sender, pan, rear main	Match by VIN after the fresh oil is traced	Oil-on-exhaust repair. Adapter and valve cover are the common ones. Rear main is the expensive miss.	TEST REQUIRED

Leave these numbers at home

NTK 23142 catalogs to other Jeeps, not this Grand Cherokee Bank 2 Sensor 2. **Mopar 56029001AB** had no fitment found. **53030006** is not a confirmed vacuum elbow for this truck. 53030008 is a PT Cruiser tee. NOT CONFIRMED

Wire repair, not a vehicle part

Broken heater or IAC wire: ratcheting crimper for heat-shrink butts, and adhesive-lined red butts, 22–16 AWG. Heat gun. Splice off the pipe. No solder sleeves, no T-taps, no tape-only joints, no grease in the O2 plug. A Weather-Pack assortment does not replace the O2 or IAC housing.

One hose is not the system

47128 does not include the booster hose, the purge line, or the HVAC harness. Replacing the short CCV pieces and leaving a split elbow at the booster still leaves the idle high. Smoke the manifold after the new pieces are on, then do the IAC and the relearn. Guides 03, 04, and 07.

Counter script. "2004 Grand Cherokee, 4.0L, this VIN. California package if I count four oxygen sensors. I need the Bank 2 Sensor 2, rear pipe, after the rear mini-cat" — or "the short CCV hoses, 4854265" — or "a new IAC, 4874373AB, and I will not let anyone push the pintle." If the counter's picture does not match the part in your hand, do not buy it.